Berry Bot

redesigning agriculture

ITIS Rossi 1878

X

SIEMENS

Istituto Tecnico Industriale Statale Alessandro Rossi

automation olympics

All big changes start small

45° 33' 9.269" N ■
11° 33' 20.269" E

A New Era for Agriculture

Our solution was born from an ambitious vision: to transform the agricultural sector through technological innovation.

Agriculture holds enormous potential for development, driven by the growing opportunities offered by modern technologies. In January 2022, the global population reached 8 billion, and by 2050 it is expected to rise to 9.7 billion. This growth is projected to significantly increase demand for agricultural products and food, putting further pressure on the land available for production.

At the same time, the agricultural workforce is shrinking dramatically worldwide, placing increasing strain on the global food system. The effects are particularly evident in fruit production, where a portion of the harvest is lost every year — enough to potentially meet European demand.

Harvesting is also a race against time. Fruit and vegetables must be picked at the right stage of maturity: delays can cause their quality and market value to deteriorate significantly.

To address these challenges, our project aims to automate strawberry cultivation and harvesting, ensuring that each fruit is picked at its optimal stage of ripeness.

BerryBot was developed to meet the needs of an increasingly uncertain agricultural sector, with benefits that extend beyond producers to the planet itself.

Nature Meets Automation

Our goal was to create a meaningful connection between automation and nature, bringing an anthropomorphic robotic arm into an unconventional setting: a greenhouse dedicated to strawberry cultivation.

We see automation as a tool that works in harmony with nature, shaping agricultural processes towards more sustainable and efficient production. BerryBot was designed to support every stage of the plant's growth, from planting to harvesting, enabling careful quality control and improving harvesting efficiency.

BerryBot is more than just a robot. It can also support plant care, protection and maintenance. Its robotic arm takes over repetitive tasks such as identifying and picking fruit, allowing us to focus on one of the most critical stages of the process: harvesting. This is achieved through a highly integrated ecosystem combining modern technologies.

Every greenhouse shelf is systematically scanned by a computer vision system, using a robot-mounted camera and a microprocessor to analyse images. To minimise waste and unnecessary movements, the system examines small sections of each shelf at a time. Once a ripe strawberry is identified, the robot picks it and transports it to a designated collection point.

Respect for nature lies at the heart of our project. BerryBot follows the plant's natural cycles while minimising stress. Harvesting takes place at night, helping reduce plant stress and potentially improve product quality.

This approach also benefits the harvesting process. Artificial lighting illuminates individual sections of the shelves, creating controlled conditions that help the computer vision system identify ripe strawberries more reliably and reduce detection errors.

A Closer Look at the World

BerryBot's interaction with its surrounding environment is made possible by a camera, which captures images of the greenhouse shelves and identifies the strawberries growing on them.

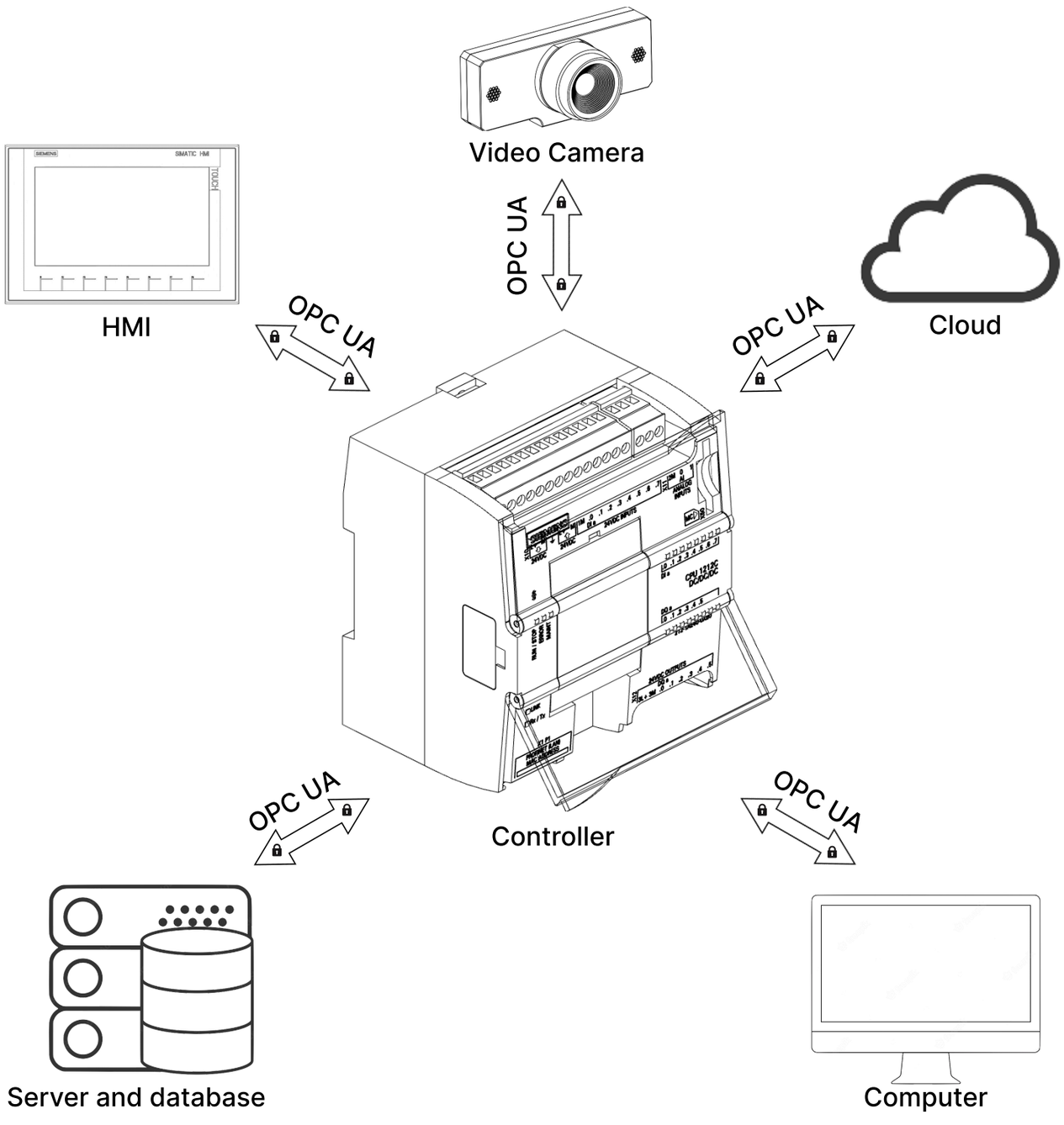
To achieve this, the system relies on computer vision running on a Raspberry Pi microcomputer, which extracts the positions of the fruit from the images and communicates them to the controller.

The image processing procedure is based on detecting the red colour of ripe strawberries, followed by calculating the coordinates of the centre of each individual fruit. The strawberry recognition and detection program does not rely on artificial intelligence, which would require multiple models to be trained and analysed in order to recognise the target object. This approach has allowed us to save processing time, making the entire process significantly faster and more efficient — a crucial factor in the harvesting operation.

The fruit positions are calculated by analysing every single pixel in the image and creating suitable masks to isolate the red colour. However, the absence of an AI system trained on multiple models can lead to potential detection errors. To address this limitation, we defined precise contours around the fruit, filtering out colour regions that are too small or fall outside the established boundaries. The algorithm also includes the calculation of the most suitable trajectory to follow when multiple strawberries are detected within the same section.

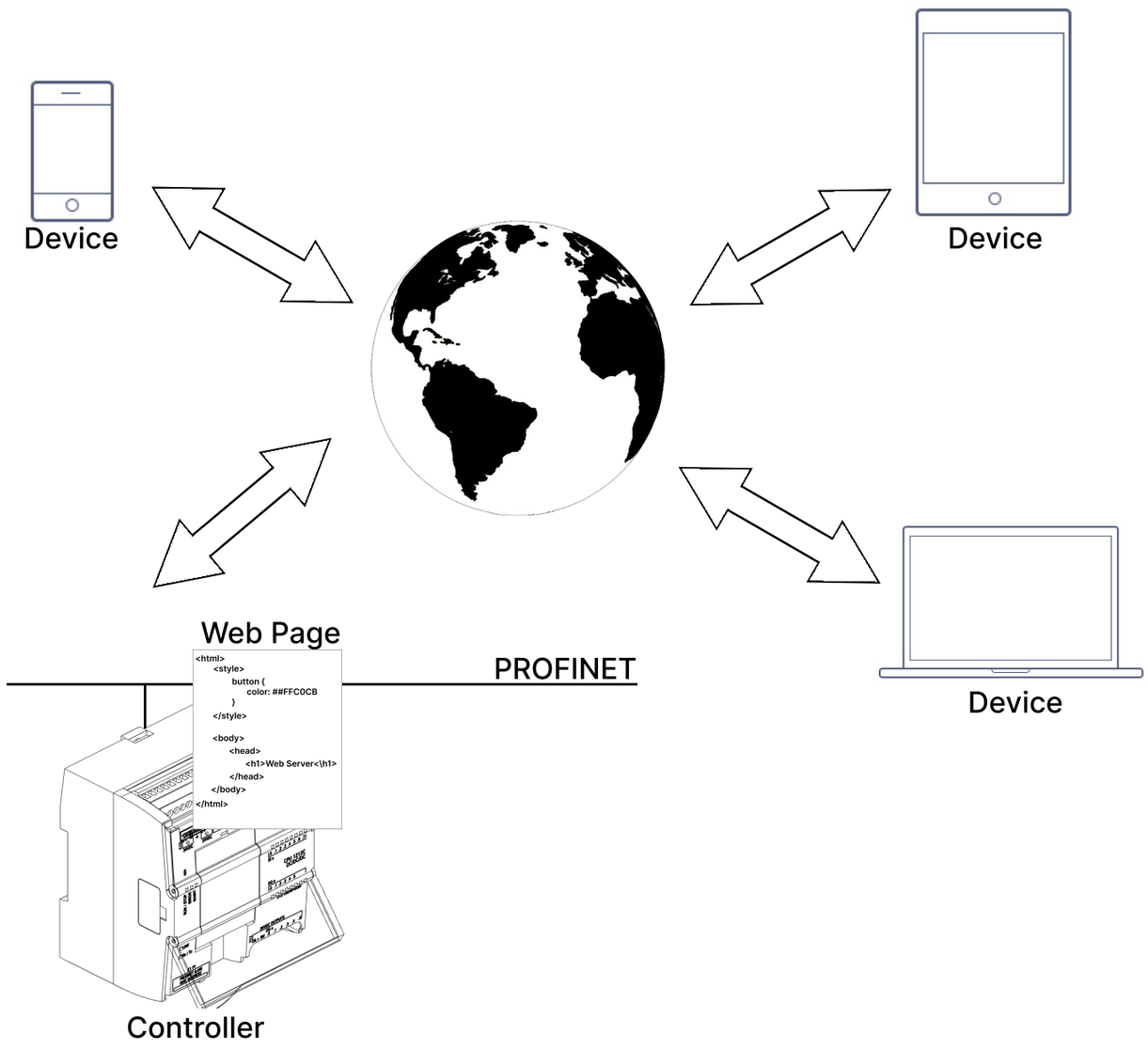
The transmission of this data to the PLC relies on an essential communication protocol that enables data exchange between the different processors, a technology underpinning Industry 4.0: the OPC UA protocol.

A key feature of this architecture is the simplicity of machine-to-machine (M2M) communication, which allows different devices to exchange data directly with one another.
However, the image analysis program cannot run directly on the controller due to incompatibilities with the specific computer vision libraries it uses.
The coordinates obtained from the camera are therefore transferred via OPC UA from the microprocessor, where the images are processed, to the PLC.
Additional variables shared through OPC UA and managed by the main controller allow the image processing program to be started and stopped as needed, ensuring synchronisation between the two processes.



Just a Click Away

A cornerstone of Industry 4.0 is the simplicity of interaction between humans and machines.
With this in mind, we designed a user interface with a clean, intuitive layout. This was made possible through careful user experience (UX) design, ensuring that aesthetics were not overlooked and that natural, effortless interaction between humans and machines remained at the heart of the experience. The result is an interface that feels familiar from the very first glance.
Interaction is seamless and always accessible, even remotely. By leveraging a web server, users can operate BerryBot and monitor the status of the process from wherever they are.
The entire interface was carefully tailored to the project. We began by designing its visual elements using a vector graphics editor, then moved on to markup and styling languages such as HTML and CSS. Finally, we tested, corrected and refined the user experience to achieve the desired result.



The Power of Integration

Seamless communication between the various controllers and microprocessors is essential, enabling data from their respective peripherals to be processed and coordinated across the entire system.

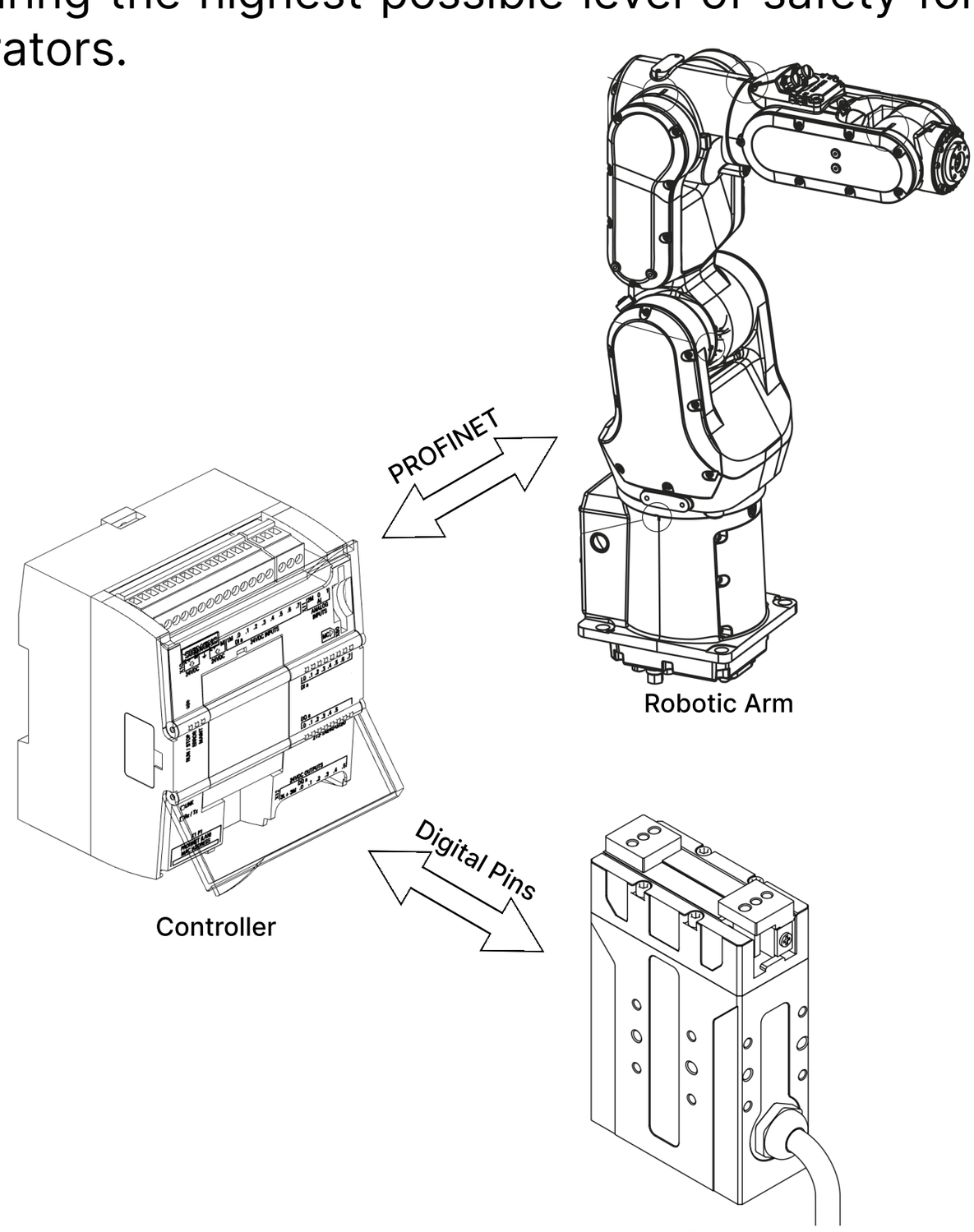
The process as a whole relies on high-level programming languages and specialised environments: Python for computer vision, RAPID for the kinematics of the anthropomorphic robot, and Structured Text and Ladder Logic for programming the Siemens PLC. The technologies are brought together through the PLC and its TIA Portal programming environment, which establishes the connection between the real-world environment captured by the camera and the mechanical operation of the robotic arm.

The coordinates obtained through the previously described image processing procedure are then sent to the S7-1200 Safety PLC using the OPC UA protocol.

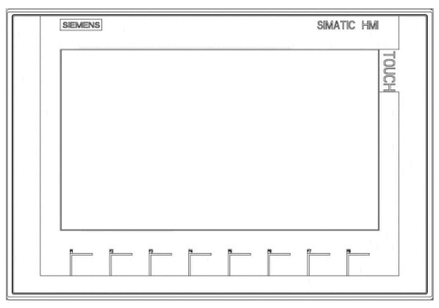
Once these coordinates have been converted to match the robotic arm's spatial coordinate system, they are transmitted from the PLC to the robot's dedicated OmniCore controller. Communication between the S7-1200 and OmniCore is established through the PROFINET protocol, with 64 bytes allocated for input and 64 bytes for output, enabling the exchange of signals and data between the two devices.

The PLC is the beating heart of the automation system. It manages signals, robot operation and communication between the various peripherals, while also playing a crucial role in ensuring safety within the robotic cell — a requirement that must be guaranteed at all times.

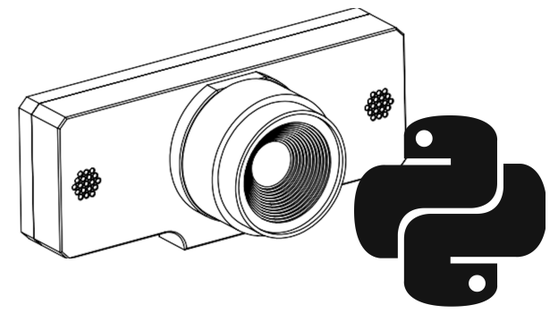
The anthropomorphic robot's controller, programmed in the high-level RAPID language using the RobotStudio platform, contains the instructions that must be executed once commands are received from the PLC.
Safety, a cornerstone of industrial automation, was a fundamental priority throughout our project. Since the anthropomorphic robot we selected is not collaborative, a dedicated safety system is required to monitor signals from light curtains and emergency stop buttons.
The Safety PLC can therefore immediately stop the robot's movements when necessary, ensuring the highest possible level of safety for operators.



The Ecosystem



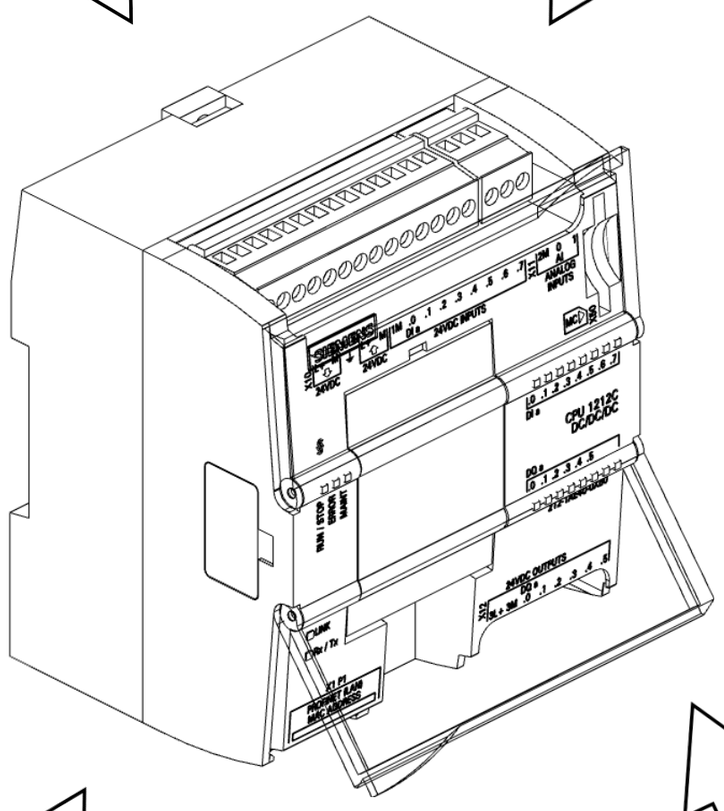
HMI



Computer Vision

OPC UA

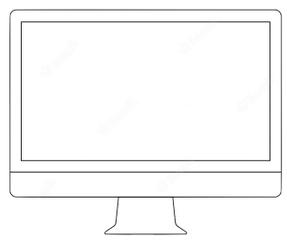
OPC UA



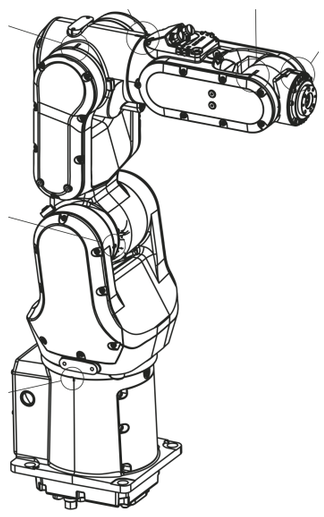
Controller S7-1200

PROFINET

Web Server



SCADA

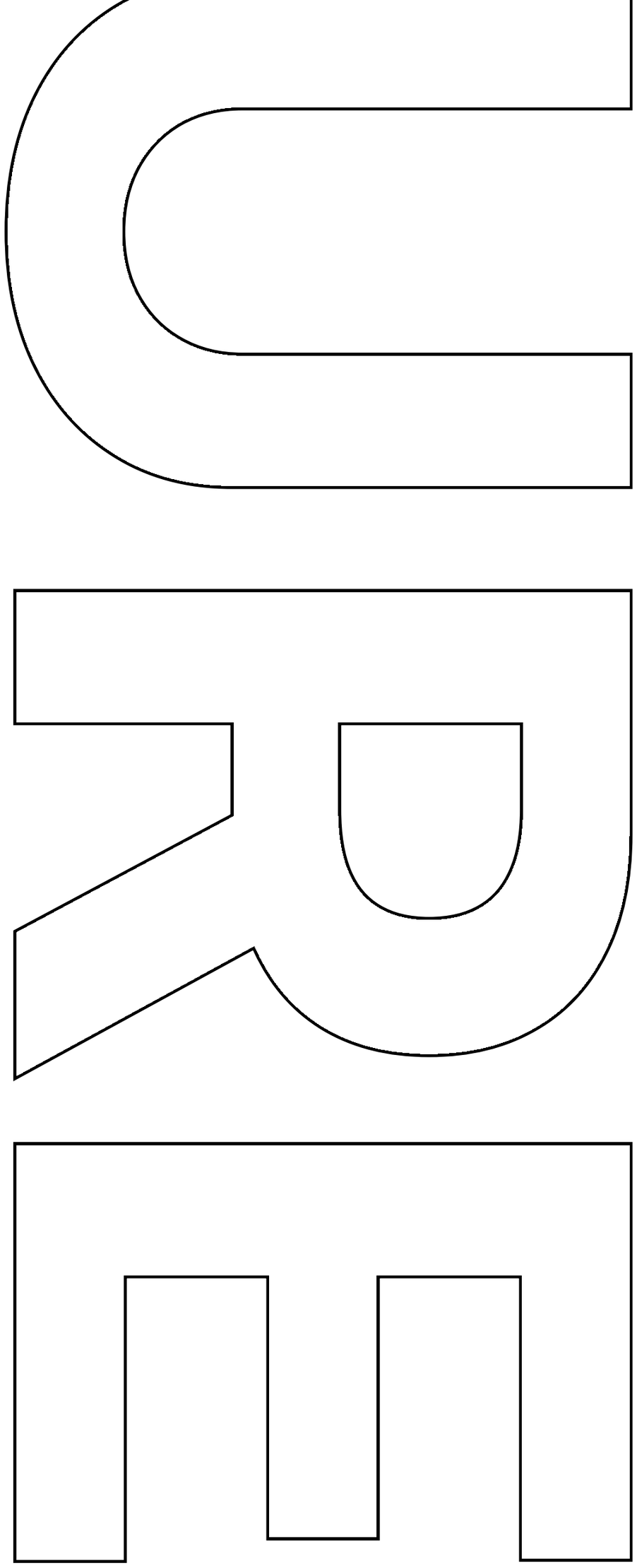


Robotic Arm

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        button {
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        }
    </style>

    <body>
        <head>
            <h1>Web Server<\h1>
        </head>
    </body>
</html>
```

Web Page



Symbiosis

Our vision is to modernise the way agriculture is practised and perceived, bringing it in line with the demands of the modern world.
We aim to create a symbiosis between humans and machines, preserving human reasoning while complementing it with the efficiency, precision and repeatability of automation.
At the same time, we strive to respect nature and its rhythms, both for ourselves and for future generations.
We believe this can be an important step towards greater automation in the agricultural sector, enabling it to meet the challenges of the future: population growth, increasing food demand and the protection of our planet.

Those Who Believed in Us

Bringing this entire automation system to life was made possible thanks to the invaluable support and collaboration of local institutions and companies.

We would like to thank our school, Istituto Tecnico Industriale Statale Alessandro Rossi, ITS Academy Meccatronico Veneto and the Parents' Committee for providing the funding that allowed us to turn our idea into reality.

We are also deeply grateful to GEVA, a leading company in metal forging automation, and to its CEO, Maicol Cavedon, for giving us the opportunity to explore new technologies, including the anthropomorphic robotic arm.

Our sincere thanks also go to Maurizio Lepori, Educational Manager at ABB, for his guidance and support in programming the robotic arm.

Last but certainly not least, we would like to thank our teachers, Gianfranco Azzolin, Giuseppe Pigatto and Alessandra Fanton, who believed in us and our idea from the very beginning. Their constant availability, encouragement and invaluable assistance played a crucial role in bringing this project to life.

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